

SAMUEL GRAYSON

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EDUCATION

NORTHWESTERN UNIVERSITY

Bachelor of Science: Computer Engineering

Dean's List Fall 2024, Winter 2025, Spring 2025, Fall 2025; High Honors Fall 2024, Fall 2025

Evanston, IL

Candidate, Jun 2027

GPA: 3.93/4.00

SKILLS

Programing Languages: C, C++, Java, JavaScript, Python, Scamper (Scheme), SQL, Racket, Verilog, SystemVerilog, VHDL

Web Development: HTML, CSS, Bootstrap, Next.js, REST API Design and Development, WebSockets, Flask, gunicorn, Nginx

Software: Autodesk Inventor, CAD, Altium, KiCad EDA, EasyEDA, Microsoft Office 365, Adobe Creative Suite, Waveforms

Other: Arduino, ESXi, Git, PlatformIO, DBEaver, TIBCO Spotfire, PostgreSQL, SQLite, ETC Eos, QLab, MagicQ, Dante

EXPERIENCE

APPLE

Platform Architecture Intern

Cupertino, CA

Incoming, Summer 2026

UNITED AIRLINES

Software Engineering Intern

Chicago, IL

Jun 2025 – Sep 2025

- Developed a full stack Flask application enabling seamless communication between flight crews and the airline's hotel operations team using Python, JavaScript, WebSockets, RESTful APIs, Twilio for SMS integration
- Implemented generative AI (LLMs) to automatically classify and route plain-text flight crew inquiries to the appropriate hotel scheduler, while performing initial data collection to streamline workflows and significantly reduce ticket resolution time

INNOX SHENZHEN

Supply Chain Intern

Shenzhen, China

Dec 2025 – Jan 2026

- Learned how to do business in China and leverage China's manufacturing ecosystem to build and prototype efficiently

VAK SUSTAINABLE COMPUTING LAB

Undergraduate Researcher

Evanston, IL

Jan 2025 - Current

- Built and programmed a multispectral imaging system using Raspberry Pis, ESP32s, and multiple cameras with spectral filters

THE ELECTRIC TUBE COMPANY

Founder

Bexley, OH

Oct 2020 - Current

- Imported vintage vacuum and nixie tube equipment from Europe and Central Asia for resale in America (eCommerce)
- Designed custom PCBs for computer-controlled nixie tube displays

BEXLEY CITY SCHOOLS

Lighting and Sound Technician

Bexley, OH

Apr 2021 - Jun 2023

- Oversaw a 10-person team and managed all lighting, sound, and video systems in Bexley City School's 3 theatrical venues
- Collaborated with vendors and stakeholders to organize >\$100,000 in capital improvements in lighting and sound equipment

PROJECTS

LEAKLOGIK VALVES

Jan 2024 - Current

- Secured \$1500 in seed funding to develop a patent pending microprocessor-controlled leak-sensing toilet shutoff valve
- Conducted product-market fit and patent research, informing design decisions and guiding strategy for IP development

NORTHWESTERN FORMULA RACING

Sep 2025 - Current

- Designed a PCB to integrate a VectorNav VN-300 IMU to the car's internal CAN system using an ESP32 and C

IEEE CHIP DESIGN TEAM (NU)

Jan 2025 - May 2025

- Designed a convolutional filter in RTL for use in the team's sign language recognition (computer vision) FPGA project

ADDITIONAL

Home Lab: Operate a variety of servers and networking equipment as well as an IBM MP3000 mainframe

Clubs: President of the Northwestern Russian Club, Resident at The Garage (Northwestern), NU Aviation Club Treasurer

Other: Received a fully funded U.S. Department of State ECA scholarship, CLS, to study Russian abroad in Kyrgyzstan (2024)